

Pythagoras, mensuration and 3D - Set 2

TMUA practice questions by topic. Specification M5.7, M5.11-M5.13, M5.15-M5.17. 5 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 From the top of a cliff 40 m high, the angle of depression of a boat is 30° . How far is the boat from the foot of the cliff?
- A 20 m
B $20\sqrt{3}$ m
C $40\sqrt{3}$ m
D 40 m
E 80 m
F $\frac{40}{\sqrt{3}}$ m
- 2 *P*: 'a solid cylinder and a solid cube have equal volumes'. *Q*: 'the cylinder and the cube have equal surface areas'. Which one of the following is true?
- A *P* is necessary and sufficient for *Q*.
B *P* is necessary but not sufficient for *Q*.
C *P* is sufficient but not necessary for *Q*.
D *P* is neither necessary nor sufficient for *Q*.
- 3 A cylinder has radius r and height h . Its volume is 54π and $h = 2r$. Find its total surface area.
- A 27π
B 45π
C 108π
D 54π
E 72π
F 36π

- 4 A square-based pyramid has base side 6 and all slant edges of length 9. Find its height.
- A $\sqrt{81 - 36}$
B $\sqrt{45}$
C 6
D $\sqrt{63}$
E $\sqrt{72}$
F $3\sqrt{6}$
- 5 The points $A(1, 2)$, $B(4, 6)$ and $C(k, 2)$ with $k > 1$ form a triangle with a right angle at B . Find k .
- A $\frac{16}{3}$
B 7
C $\frac{28}{3}$
D $\frac{25}{3}$
E 5
F 8